# PigeonLink: Infrastructure for the Agentic Future

**White Paper**

**Version**: 1.0 | May 2025

**Authors**: Vidhi Kothari (Co-founder & CEO) And Rajpreet Singh (Co-founder & CTO)

# TABLE OF CONTENTS

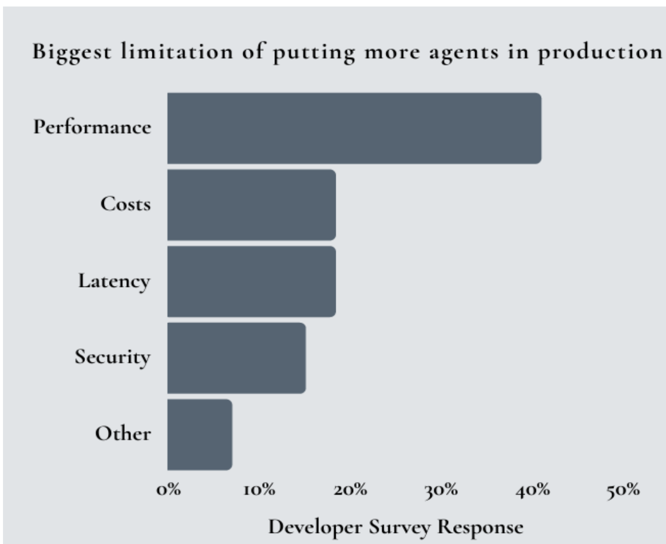
# 1. ABSTRACT

The rapid proliferation of AI agents across industries is bottlenecked by infrastructural complexity, extreme scaling costs, and unreliable inter-agent communication. **PigeonLink** introduces a infrastructure-centric solution that abstracts AI infrastructure away from developers while enabling high-performance deployment, agent orchestration, and real-time inter-agent communication. With optimized hardware layers, distributed inference, and a programmable agent meta-language, **PigeonLink** positions itself as the foundation of the future Agentic Internet and Agentic Enterprise.

# 2. INTRODUCTION

AI agents—modular, semi-autonomous software units—are revolutionizing workflows in finance, manufacturing, telecom, and software development. However, enterprises encounter severe bottlenecks in deploying and scaling multi-agent systems. **PigeonLink** addresses these challenges by offering a cloud-native infrastructure designed specifically for agent programming, deployment and coordination with full control and flexibility over their complex behavior. This enables more refined reliability analysis, observability, and verification of these "agentic programs". The key problems identified are as follows:

- **Unsustainable Scaling Costs**: AI infrastructure costs projected to hit $421B by 2033.
- **Poor Coordination**: 60% of projects fail due to lack of standardized deployment protocols.
- **Communication Failures**: Agent collaboration breaks down due to lack of shared protocols.
- **Debugging Nightmares**: Developers struggle to trace requests across agent networks.




# 3. ARCHITECTURE OVERVIEW

## 3.1Infrastructure Stack:

- **Optimized Hardware Layer**: Custom kernels and OS extensions for high-throughput, low-latency AI inference.
- **Distributed Inference Engine**: Dynamically balances agent workloads across nodes with automatic scaling and fallback mechanisms.
- **Communication Protocol**: A lightweight agent-to-agent protocol for asynchronous and real-time messaging.

## 3.2 PigeonLink's own Meta-Language:

PigeonLink introduces a **domain-specific language (DSL)** for defining agent behavior, workflows, and inter-agent relationships allowing for more fine-grained control.

- Expressive syntax for defining triggers, workflows, and constraints.
- Agents can be defined declaratively and updated at runtime.
- Built-in primitives for state, memory, and goal tracking.

## 3.3 Agent Network Topology:

PigeonLink supports both **centralized** and **peer-to-peer** agent deployments with:

- Locality-aware message routing.
- Priority queues and dependency resolution.
- Sandboxed agent execution with audit trails for observability.

# 4. USE-CASE 1: INDUSTRIAL AUTOMATION

**Scenario**: A factory IT team wants to automate quality inspection, predictive maintenance, and inventory tracking.

**How PigeonLink Helps:**

- Agents for visual inspection, fault prediction, and stock monitoring deployed via one-click provisioning.
- Scalable runtime auto-adjusts to factory floor workload.
- Agents collaborate via PigeonLink's P2P protocols to adjust schedules, reroute inventory flows, and flag anomalies in real-time.

**Benefits:**

- Zero manual scaling.
- Real-time coordination between task-specific agents.
- Minimal infrastructure overhead for IT teams.

# 5. USE-CASE 2: AGENTIC INTERNET

**Scenario:**

A user visits a complex financial services website looking for tailored advice on tax-saving investment options, risk profiling, and account setup—all in one session.

### *How It Works with PigeonLink:*

Each **webpage is dynamically supported by a swarm of specialized agents**, running on PigeonLink's infrastructure:

- **Intent Parsing Agent**: Intercepts the user's query and determines sub-intents (e.g., "risk profile", "investment comparison").
- **Finance Analyst Agent**: Retrieves market data and simulates tax scenarios using the user's profile.
- **Document Summarizer Agent**: Parses policy documents and extracts relevant clauses.
- **Dialogue Coordinator Agent**: Orchestrates replies and ensures consistency in tone, scope, and context.
- **UI Proxy Agent**: Translates multi-agent outputs into real-time web interactions—modals, tooltips, and dynamic form suggestions.

All agents **communicate over PigeonLink's inter-agent protocol** and share memory using the distributed state store. The system ensures:

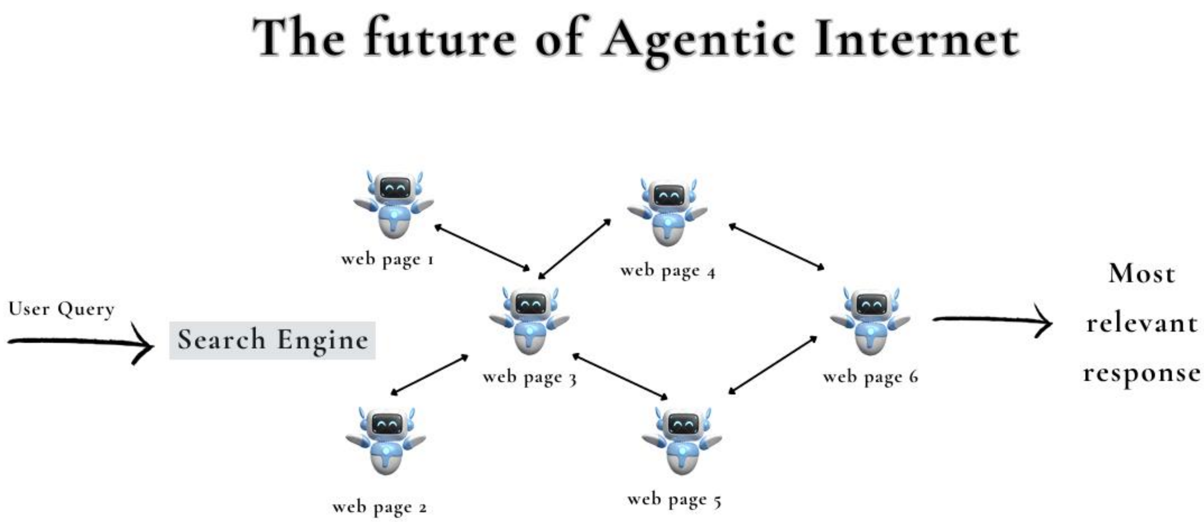
- Latency under 100ms per interaction.
- Seamless failover if any agent fails.
- Full traceability for debugging.

### *User Experience:*

The user receives a **conversational, personalized web interface**, with answers sourced in real time from multiple coordinated agents. The backend is invisible—only the intelligence is felt.

### *Benefits:*

- Reduces bounce rate by 45% through hyper-personalization.
- Eliminates backend boilerplate for developers—agents handle logic.
- Site owners get analytics on agent reasoning paths, enabling continuous optimization.




This is more powerful when agents from different websites communicate to serve better information retrieval. For example, **Netflix-IMDB-Rotten Tomatoes** and agents from these websites help users to decide the next movie to watch.

# 6. DIFFERENTATION & MARKET POSITIONING

## *Why PIGEON Stands Out:*

- **Integrated Kernel Support**: Unlike traditional agent runtimes, PigeonLink provides OS-level hooks for faster context switching and shared memory pools with a new pipelining mechanism.
- **End-to-End Observability**: Every agent interaction is logged and visualizable.
- **Developer Experience**: Think "Vercel for Agents" — one-command deployments, intuitive agent scripting, and automatic routing.

## *Market Readiness:*

- 93% of IT leaders plan autonomous agent initiatives.
- 86% of enterprises need tech stack upgrades to deploy agents effectively.

# 7. PRICING AND BUSINESS MODEL

- **Usage-Based Pricing**: Pay only for the compute and coordination resources consumed.
- **Freemium Tier**: Basic agent runtime for start-ups and small-scale experimentation.
- **Enterprise Tier**: SLA-backed deployments with private cloud support and advanced observability.

# 8. ROADMAP

Q1 2026: Beta access to first 100 developers.

Q3 2026: Full commercial launch.

# 9. TEAM

- **Vidhi Kothari (CEO)**: Former investor with deep GTM experience, previously at Morgan Stanley and Copia Automation.
- **Rajpreet Singh (CTO)**: MSc in HPC from TU Munich and intern at Bell Labs, with research expertise in distributed systems and autonomous workflows.

# 10. CONCLUSION

As AI transitions from monolithic models to collaborative agent-based systems, the need for robust infrastructure becomes critical. **PigeonLink** bridges the gap by offering a unified platform that handles deployment, communication, and scaling — making it the operating layer of the Agentic Future.